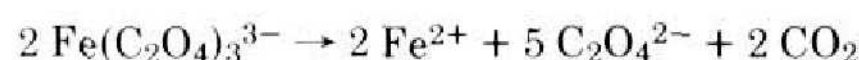


The Ferrioxalate Actinometer: A Lecture Demonstration

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A photochemical reaction can be demonstrated most strikingly by filling a 10 in. test tube with potassium ferrioxalate actinometer solution (1) and discharging a photographic strobe¹ adjacent to it. The greenish solution turns red immediately where light has been absorbed. The quantum yield for this reaction is about 1.2 at 365 nm (2), and the complex mechanism can be simplified to the overall reaction represented by the equation (3)



The free ferrous ion formed then reacts to form the red phenanthroline complex, and the absorbance of the solution is a measure of the quantity of light absorbed by the actinometer.

The actinometer solution is prepared easily^{2,3} by weighing 0.2–0.4 g of $\text{K}_3\text{Fe}(\text{C}_2\text{O}_4)_3$ into a test tube and adding about 30 ml each of a buffer (600 ml of 1 M sodium acetate and 360 ml of 0.5 M sulfuric acid diluted to 1 l) and a 0.2% aqueous solution of 1,10-phenanthroline. The potassium trisoxalatoiron(III) may be replaced by solutions of iron(III) sulfate or nitrate and potassium oxalate mixed just before use (6).⁴

The remarkable change that occurs as a result of irradiation recommends this as a demonstration of photochemistry. It illustrates the primary process in blueprinting (7) and photolytic redox reactions (charge transfer to metal) in general. A reasonably powerful strobe may also be used to initiate the $\text{H}_2 + \text{Cl}_2 \rightarrow 2 \text{HCl}$ reaction in a balloon that has been carefully purged of oxygen.

Literature Cited

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¹ A Vivitar® 281 strobe was used. It has a light output of 2800 BCPS (Beam Candle Power Seconds) and flash duration of 0.001 sec. The guide numbers for this strobe are 60 for ASA 25 and 150 for ASA 160 film. The checker reports that a minimum of 100 Watt-seconds or 2000 BCPS is required.

² Potassium ferric oxalate is available from Pfaltz and Bauer, Inc., 375 Fairfield Ave., Stamford, CT 06902 (#P22950) or from Ventron Corporation—Alfa Products, 42 Congress St., Beverly, MA 01915.

³ The preparation of this compound is presented as an exercise in several general chemistry laboratory manuals, or may be found in references (4) and (5).

⁴ Dissolve 1 g of $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ in 100 ml of distilled water to which 2 ml of concentrated nitric acid has been added to retard hydrolysis. To 10 ml of this solution, add about 60 ml of 0.1 M potassium oxalate and 10 ml of 1,10-phenanthroline solution.

Silver Mirror

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A dramatic demonstration suitable for inorganic and organic chemistry classes, as well as for general audiences, is the production of a silver mirror. The demonstration can be used to show the reduction of a metal salt to the metal, the reducing power of sugars, or both as the occasion may require. The plating of a silver mirror has been described before.¹ The following procedure based on the work of Gupta, et al.,² produces excellent mirrors and the solutions are stable over a long period of time. In addition, a change from a previous procedure³ keeps the AgNO_3 and the NH_3 solutions separate until time for use. This lowers the chance of the formation of possible explosive ammoniacal solutions of silver.

Demonstration

A scrupulously clean flask, jar, or pop bottle, and the following solutions are required:

Solution A: 2.5 g glucose and 2.5 g levulose (or 5 g honey) in 50 mL water. Add 0.6 g tartaric acid, boil, then cool the solution. Add 10 mL ethanol to stabilize and dilute to 100 mL.

Solution B: 4.0 g silver nitrate in 50 mL water

Solution C: 6.0 g ammonium nitrate in 50 mL water

Solution D: 10 g sodium hydroxide in 100 mL water

Into the clean container pour 10 mL A. Mix 5 mL B with 5 mL C and add to the flask. Quickly pour in 10 mL D, stopper and mix. Within a few seconds all surfaces of the container must be wetted with the solution for a good plating. Continue to rotate and tilt the container to keep all surfaces wet. In about one minute a fine mirror is formed (if the flask was clean).

Immediately pour the residue into a waste sink with plenty of water and rinse the flask thoroughly to prevent possible explosive mixture formation.

Tested Demonstrations is a monthly feature designed to present lecture demonstrations and experiments in a format convenient for classroom use. Readers interested in either submitting or checking demonstrations should contact the column editor. An outline of format requirements was given on p. 166 of the March 1976 issue of THIS JOURNAL.

George Gilbert's interest in chemistry demonstrations traces from Hanover High School (Mass.) through undergraduate work at Antioch College (BS 1958) and his thesis at Michigan State University (PhD 1963). Additional areas of interest in chemical education include self-pacing, TIPS, and project-style laboratories. He consults with science museums and is active in fostering science-awareness in public arenas. Recreational interests include canoe-tripping, backpacking and outdoor activities. He suffers from frequent attacks of paronomasia which is best treated by studied forbearance.

